SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 09:45 UTC

SEC P vs NP — daily log 2026-04-28

6 cycles recorded

Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) × Boolean formula depth lower bounds for KRW-style block compositions $f \circ g$, the central complexity-theoretic object behind $P \not\subset NC^1$; equivalently the depth complexity $D(f \circ g)$ of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded**: 2026-04-28 01:39 UTC
- **Entry ID**: `8047f7f8eea1`

Statement

For all boolean functions $f:\{0,1\}^n\to\{0,1\}$ and $g:\{0,1\}^m\to\{0,1\}$ for which $HX^1(X_f)$ and $HX^1(X_g)$ admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace_pairing on the KW-metric spaces obeys the sub-additive bound $\kappa(f \circ g) \geq \kappa(f) + \kappa(g) - C$, where C is a universal constant independent of n,m, and where $(X_{f\circ g}, d_{f\circ g})$ is built by the coarse_product operation. In particular, witness cocycles $c_f \boxtimes c_g \in HX^1(X_{f\circ g})$ and tensor Roe operators $T_f \otimes T_g \in C_u^*[X_{f\circ g}]$ realize $\log|\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle| / \log(\mathrm{propagation}(T_f \otimes T_g)) \geq \kappa(f) + \kappa(g) - C$.

Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 (κ ≤ depth) to recover the KRW conjecture $D(f \circ g) \geq D(f) + D(g) - O(1)$. The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: $\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle = \langle c_f, T_f\rangle \cdot \langle c_g, T_g\rangle$ (Connes-style product), while the propagation of $T_f \otimes T_g$ equals $\mathrm{propagation}(T_f) \cdot \mathrm{propagation}(T_g)$ under the depth-additive metric of coarse_product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has κ = O(1), so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse_product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
         ^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) at build_X_d before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix build_X_d to enumerate vertices correctly (e.g., X = [tuple(b) for b in itertools.product([0,1], repeat=n)]) and rerun the 25-pair × 5-seed protocol.

Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity × Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. AC^0[⊕] regime, expressed via the coarse 1-cohomology HX^1(X_f) and the Roe-trace pairing on C_u^*[X_f]
- **Recorded**: 2026-04-28 03:39 UTC
- **Entry ID**: `264c40c0d040`

Statement

Let f_n = ⊕-tree_n be the balanced binary tree of XOR gates on n = 2^k inputs, with KW-metric space (X_{f_n}, d_{f_n}). Define the coboundary δ : C^0_coarse(X_{f_n}) -> C^1_coarse(X_{f_n}) and the propagation-R Roe subalgebra C_u^*[X_{f_n}]_R. Then there exists a universal constant α ∈ (0,1) such that, for all R ≤ α·log_2 n, every coarse 1-cocycle c with propagation ≤ R is a coboundary (so its class in HX^1 is trivial), while for R ≥ k = log_2 n there exist nontrivial classes c whose Roe-trace pairing satisfies log|⟨c,T⟩| / log(propagation(T)) ≥ k − O(1). Consequently κ(f_n) ≥ log_2 n − O(1), matching depth(f_n) = Θ(log n) up to constants and certifying axiom A1 sharply for an explicit recursive function.

Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known (depth(⊕-tree_n) = log_2 n).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation R ≈ log_2 n, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that κ is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of ℓ_p uniform Roe algebras

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on R*(n), dim HX^1_R, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across n ∈ {4,8,16,32}.

Multiplicity-Doubling Lower Bound from Protocol Pullback on Indexing Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Roe-Algebra methods in Communication Complexity × Deterministic two-party communication complexity CC(f∘G^n) of lifted boolean functions
- **Recorded**: 2026-04-28 08:37 UTC
- **Entry ID**: `6a6a970e1f5b`

Statement

Let G = IND_b be the standard indexing MetricGadget on $X = \{0,1\}^b \times [b]$, $Y = [b]$ with d the Hamming metric on X×Y, and let $f: \{0,1\}^n \to \{0,1\}$ have deterministic decision-tree complexity Q(f). Then for the lifted input space $((X\times Y)^n, d_\oplus)$ at scale R = diam(G), every deterministic protocol Π computing $f\circ G^n$ satisfies CoarsePullbackProtocol(Π) = (m_Π, R_Π) with $m_\Pi \geq 2^{Q(f)} / \mathrm{poly}(b, n)$, so by axiom A2, $CC(f\circ G^n) \geq Q(f) - O(\log(bn))$. Equivalently, CLC(f, IND_b) ≥ Q(f) · $\log_2(\mathrm{asdim}_R(\mathrm{IND}_b^{\otimes n})+1) - O(\log bn)$, recovering the Raz–McKenzie bound through the coarse-cover route.

Rationale

This sub-conjecture directly tests axiom A2 (Protocol → Cover) on the canonical gadget for which Raz–McKenzie lifting is known to hold, using axiom A3 (Asdim Amplification) to guarantee that the iterated indexing gadget has unbounded coarse dimension proportional to n. If A2 is sound, the multiplicity of the protocol-induced RoeCover must scale exponentially in the query complexity, otherwise one could collapse a query-tree lower bound through a low-multiplicity cover — contradicting the structure theorem at the heart of CGL. Confirming the bound on small parameters provides empirical evidence that the coarse-cover viewpoint reproduces the standard query-to-communication lift, a necessary sanity check before pursuing axiom A4's non-relativizing strengthening.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2012.03415v1] On Query-to-Communication Lifting for Adversary Bounds

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38168056.py", line 114, in <module>
    seeds = [int(x) if x else 11 for x in input().split()]
                                          ^^^^^^^
EOFError: EOF when reading a line
```

Judge reasoning

The test crashed with an EOFError while reading seeds from stdin before any data could be produced, so neither the support nor falsification conditions can be evaluated. | next: Patch the harness to provide default seeds (or read from argv/env) when stdin is empty, then rerun the ≥20-trial battery over b∈{2,3,4}, n∈{2,3,4}.

Σ^b_0-PIND Width-2 Closure Round-Count vs DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic / Cook-Reckhow-Krajicek correspondence: the Σ^b_0-PIND round count $r_2(F)$, defined as the minimum $k \in \mathbb{N} \cup \{\infty\}$ such that iterating the closure operator $\Phi(S) = S \cup \{C : |C| \leq 2$ and C is RUP-derivable from $F \cup S$ in one unit-propagation pass$\}$ k times yields $\bot \in \Phi^k(\emptyset)$; r_2 captures the depth of sharply-bounded polynomial-time

induction in PV restricted to width-2 reasoning (Cook-Reckhow lineage; rarely measured directly as a quantitative invariant on small unsatisfiable CNFs). × Lex-DPLL search-tree leaf count L(F) on small unsatisfiable 3-CNF formulas under a fixed lexicographic variable order with unit propagation, equivalently tree-resolution leaf count up to constants.
- **Recorded**: 2026-04-28 08:46 UTC
- **Entry ID**: `fc9124ecdaa8`

Statement

For every unsatisfiable 3-CNF F on $n \leq 14$ variables and $m \leq 6n$ clauses: (a) if $r\_2(F) < \infty$, then $\log_2 L(F) \leq 2 \cdot r\_2(F) + 2 \cdot \log_2(n+1)$; (b) if $r\_2(F) = \infty$, then $\log_2 L(F) \geq n/4$. Equivalently, finiteness of width-2 RUP closure forces a polynomial DPLL bound, while non-closure forces an exponential one — a sharp dichotomy.

Rationale

Width-2 RUP closure is the propositional shadow of Cook's PV restricted to the sharply-bounded Σ^b_0-PIND fragment, the weakest layer of the Buss hierarchy. The Cook-Reckhow-Krajicek correspondence predicts a polynomial relation between sharply-bounded PV-proof depth and tree-like resolution size; quantifying it should expose the exact slack and yield a dichotomy mirroring the classical PHP/Tseitin obstructions to weak proof systems.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [2407.17947v2] Supercritical Size-Width Tree-Like Resolution Trade-Offs for Graph Isomorphism - [1502.02131v2] Extracting verified decision procedures: DPLL and Resolution - [1411.7087v6] Consistency proof of a fragment of PV with substitution in bounded arithmetic - [2306.08535v1] Cyclic proofs for arithmetical inductive definitions

Empirical Test

- exit code: `124`, elapsed: `180.07s`

```
TIMEOUT after 180s
```

Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support nor falsification conditions could be evaluated. The critic also challenges, and no instances were verified. | next: Reduce the search space (e.g., cap n at 10 and reduce seeds per (n,α) pair) or parallelize/optimize the r_2 closure and DPLL leaf-count routines so the test completes within the time budget.

Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF_min

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Number theory of polynomial heights — the (logarithmic) Mahler measure m(p) = log|a_d| + sum_{|α|>1} log|α| over the complex roots of p ∈ Z[x] (Lehmer-Boyd-Smyth tradition: Lehmer's problem on smallest non-cyclotomic measure, Boyd's heights, Smyth's resolved cubic case). Rarely deployed in meta-complexity; computable in poly time via numerical root-finding on the explicit degree-(2^n − 1) integer polynomial. × DNF_min(f): the minimum number of product terms in a DNF representation of a Boolean function f:{0,1}^n→{0,1} given its

2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy at the heart of meta-complexity worst-case-to-average-case reductions.
- **Recorded**: 2026-04-28 09:18 UTC
- **Entry ID**: `a6aec390c941`

Statement

Define the truth-table polynomial p_f(x) := sum_{i=0}$^{\{2}$n − 1} f(i)·x^i ∈ Z[x], where f(i) is the value of f on the n-bit binary expansion of i. For every f with |f^{-1}(1)| ≥ 1, m(p_f) ≤ (DNF_min(f) − 1) · log_2(n + 1). Equivalently, every f whose truth-table polynomial has logarithmic Mahler measure exceeding (k − 1)·log_2(n+1) satisfies DNF_min(f) > k.

Rationale

Each DNF product term corresponds to a coordinate-aligned subcube whose index set is a coset a+W ≤ F_2^n; its indicator polynomial factors as x^a · prod_j (1 + x$^{\{2}${i_j}}), so all roots lie on the unit circle and m = 0 (purely cyclotomic). Disjunction of k subcubes makes p_f a signed sum of k such cyclotomic-rooted polynomials, and Boyd-style sub-additive heuristics for Mahler measure under integer-coefficient sums predict at most a log_2(deg+1) escape per added term. If true, this transforms a hard Diophantine height into a poly-time-computable lower bound on DNF_min, giving a Lehmer-type obstruction useful in restricted-MCSP regimes.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

Plünnecke-Ruzsa Doubling of Constraint Vectors Lower-Bounds SoS Degree for 3-XOR

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics in F_2^n: the Plünnecke-Ruzsa doubling constant K(A)=|A+A|/|A| and its log version σ(A)=log_2|A+A|-log_2|A| of a finite set A⊆F_2^n (Freiman-Ruzsa / Bogolyubov / Sanders’ polynomial Bogolyubov tradition). The mathematical branch is harmonic / arithmetic combinatorics on abelian groups, distinct from polyhedral / Newton-polytope geometry, and rarely deployed in SoS lower bounds for CSPs. × Sum-of-Squares refutation degree d_SoS(F) of unsatisfiable 3-XOR instances F over GF(2), accessed via the Grigoriev-Schoenebeck combinatorial proxy g*(F) = min{|T| : ⊕*{i∈T} a_i = 0 ∈ F_2^n and* ⊕{i∈T} b_i = 1 ∈ F_2}, which sandwiches SoS degree to within ±1 by the Schoenebeck (2008) characterization for 3-XOR.
- **Recorded**: 2026-04-28 09:45 UTC
- **Entry ID**: `6ae7590ec4aa`

Statement

For every unsatisfiable 3-XOR instance $F = ((a_i,b_i))_{i=1..m}$ on $n \le 20$ variables with constraint multiset $A = \{a_1,\dots,a_m\} \subseteq F_2^n$, the Schoenebeck combinatorial SoS-degree satisfies $g(F) \ge \sigma(A) + 1$, where $\sigma(A) = \log_2|A+A| - \log_2|set(A)|$. Equivalently, $|A+A| \le |set(A)| \cdot 2^{g(F)-1}$. A single (F, A) violating this inequality refutes the conjecture.

Rationale

Schoenebeck's theorem says small SoS degree forces a short additive dependency among constraint vectors; Plünnecke-Ruzsa says short additive dependencies push A toward an affine-subspace structure with low doubling. The bridge predicts a monotone, never-before-quantified link between structurelessness (large doubling) of the constraint vectors and the unavoidable length of a Schoenebeck certificate, transferring Freiman-Ruzsa intuition into proof complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2507.01005v1] Discovery and study of the eclipsing variable star Grigoriev 1 - [0701009v2] Electron mobility on a surface of dielectric media: influence of surface level atoms - [1812.10216v2] Robust Approach for Rotor Mapping in Cardiac Tissue - [1701.04521v1] Sum of squares lower bounds for refuting any CSP - [0305179v3] Polynomial Degree and Lower Bounds in Quantum Complexity: Collision and Element Distinctness with Small Range

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 70, in run_trial
    set_A_size, A_plus_A_size, sigma_value = sigma(A)
                                             ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 56, in sigma
    A_plus_A.add(xor(a, b))
                 ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 21, in xor
    return x ^ y
           ~~^~~
TypeError: unsupported operand type(s) for ^: 'list' and 'list'
```

Judge reasoning

The test crashed with a TypeError in the xor() helper (xoring two lists), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the xor() helper to operate on integer bitmask representations of F_2^n vectors (or convert lists via tuple/int) and re-run the 3000-instance sweep.